

Tel Aviv University
The Iby and Aladar Fleischman Faculty of
Engineering

FPGA Laboratory

Experiment FPGA 3 – 7 Segment and LEDs

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Lab Goals

The purpose of this lab is to cover the following topics:

1. FPGA Design Flow in Xilinx Environment:
 - a. Design and implement a project on a basic FPGA board
 - b. Interfacing the peripheral components of the BASYS3 board
 - i. Utilize the 7-segment display and LEDs
 - ii. Understand the constraints file (.xdc)
 - c. Run and understand Synthesis and Implementation
 - d. FPGA device programming

Submission Guidelines

The lab consists of TBD tasks, where each task depends on the previous.

1. Submit a report (PDF format), as well as the XDC constraints file, and all the *.v files you have edited. All zipped together in a zip file named FPGA_3 <ID1> <ID2>
2. Each waveform must be annotated by (i) drawing arrows that point to important signals and (ii) explaining these signals.
3. Most of the modules in this lab are parametric (for example, Compadder(N) has a design parameter N, which determines the bit-width of its operands). These modules must in fact support different parameter values, and not to stick to some constant value.
4. Each Verilog source/test-bench header must have both of your names in it:

```

1 `timescale 1ns/10ps
2 //////////////////////////////////////
3 // Company:      Tel Aviv University
4 // Engineer:     Your name
5 //

```

5. At home you can work with the [Vivado Web-Pack](#). You will need to create a free student account in Xilinx before downloading and installing the software.
6. It is your responsibility to either use a USB-flash drive or other cloud solutions to save your project before you leave the laboratory. The laboratory PCs are erased from time to time.

1. Create new Vivado project

Launch the Vivado software, create a new project. Follow the “Vivado Project Creation” subsection of the Vivado-tutorial to set up a new project. The name of the project should be **lab3**, and it should be saved in **G:\FPGA_LAB** path. The project type is **RTL Project**. You may add the provided Verilog sources as well as lab3_constraints.xdc constraints-file. Make sure to choose the **xc7a35t1cpg236-2L** part in the “Default Part” screen of the wizard. Notes:

- To add more sources later, go to “Project Manager□Add Sources” in the flow-navigator in the left part of the Vivado screen.
- For this project you will need to use sources from previous designs: FA.v, Compadder.v, Limited_incremementor.v and counter.v

- In this lab, all the tasks are to be performed under the same “lab3” project.

2. Seven Segment Display

The 7-segment digits are refreshed periodically with one digit lit at a time for a specific period. The total refresh time of all 4 digits must be between 16ms and 1ms for a human eye to see the display without the blinking effect. Consider the provided Seg_7_Display module.

```
module Seg_7_Display(
    input [15:0] x,
    input clk,
    input clr,
    output reg [6:0] a_to_g,
    output reg [3:0] an,
    output wire dp
);
```

- Add the provided design source Seg_7_Display.v to your project.
- Adapt the code of Seg7_display module to the appropriate refresh frequency by selecting the correct bits of the clkdiv register (line 46).
- In the report: describe your choice and justify your answer analytically.

3. Display Playground

The display playground is a top module that utilizes the 7-segment display to display the values selected by a certain logic. There are 16 switches in a row on the board, we divide them into 4 groups of 4 switches (SW12-SW15, SW8-SW11, SW4-SW7, SW0-SW3), each group indicates a digit in the display. In addition, it controls the LEDs (0-9) to indicate the first digit on the display. For example, if the selected switches are 0001001000000011, the 7-segment display will show 1203 and LD3 will be on. You can choose a reasonable behavior for the case where switches do not indicate a digit. Consider the provided Display_Playground module.

```
module Display_Playground(
    input clk,
    input [15:0] sw;
    output wire [6:0] seg;
    output wire [3:0] an;
    output wire dp;
    output reg [9:0] led;
);
```

- Add the provided design source Display_Playground.v to your project.
- Design a block diagram for the display playground module, use the Seg_7_Display module as an instantiated module (No need to show the internal logic), show and explain any other logic necessary.

- c. What is the behavior you chose for the case where switches do not indicate a digit? Explain what you did.
- d. Implement a module by filling the Display_Playground.v Verilog source according to your block diagram.
- e. Set Display_Playground as a Top module under the “Design Sources” folder.
- f. In the flow navigator, RTL analysis, open the elaborated design schematic, make sure that the schematic seems correct and no inputs or outputs are missing. Provide the screenshot of the elaborated design schematic.
- g. Synthesize the design (In the flow navigator, go to Synthesis → Run Synthesis), then open synthesized design schematic and describe the differences that appeared compared to the elaborated design (previous item).
- h. Read the “Messages” tab (near Tcl Console, in the bottom of Vivado window) and fix your design until no critical warnings or errors remain.

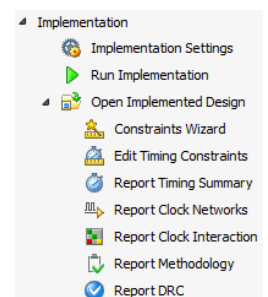
4. Setting up constraints

The constraints file defines the clock pin, clock constraints, and the IO your design will interface the board with.

- a. Open the lab3_constraints.xdc file, make sure it is added to your project.
- b. Set up the clock signal input by uncommenting the 2 lines that connect the clk signal of your top module to the W5 pin of the FPGA.
- c. Provide a timing constraint by uncommenting the line that begins with create_clock command
- d. Set up the input IO pads by uncommenting the lines that interface the inputs of your design.
- e. Set up the output IO pads by uncommenting the lines that interface the "seg" and the "an" buses, as well as the "dp" output line (the output that is in charge of the dot nearby each one of the 7-seg digits). In addition, pick ten led output pins (lines 65-76 of the constraints file) and map them to led[9:0] ports.

5. Implementing the design

- a. In the flow navigator, go to Implementation → Run Implementation
- b. After implementation is complete, check (again) for errors and critical warnings.
- c. You should notice an ILA debug core appearing in your synthesized design block diagram view. Provide a screenshot of your synthesized design schematic.



- d. Provide a screenshot of the design timing summary where you must exhibit positive slack both for setup and for hold times.
- e. Specify the lengths (in nanoseconds) of your critical setup path (i.e. *path length* with the lowest slack, don't just specify the slack value).

6. FPGA programming

Warning: Do not proceed to FPGA programming if the implementation procedure ended with critical warnings or errors as it might damage the FPGA device!

In this part you will work with the final stage of Vivado's Flow Navigator, called "Program and Debug"

- a. Follow the three steps from the "Programming the FPGA" section in the Vivado-tutorial (1-bitstream, 2-connect, 3-program via Jtag).
Pay attention to the correct orientation of the USB cable as shown:
- b. You need to make sure you see a correct functioning of your design. In the final meeting at the lab, you will need to show an instructor the working design. Make sure to prepare it at home in advance. You won't have time to fix the design at the lab.
- c. After your design was successfully approved, **remove the debug core** (see Vivado-tutorial document), re-implement the design. Provide a screenshot of your Project Manager → Project Summary screen including post-implementation Timing, Power, Utilization details (table view).

